UPPER RESPIRATORY TRACT INFECTIONS
Otitis Externa – Acute and Chronic

Inflammation of the skin and sub dermis of the external ear canal, also known as 'swimmer's ear' or 'tropical ear'.

Acute otitis externa - inflammation of less than 6 weeks duration, typically caused by bacterial infection.
Chronic otitis externa - inflammation of longer than 3 months, typically caused by allergies or inflammatory dermatological conditions, infective causes include fungal infection.
Necrotising (malignant) otitis externa - potentially life-threatening progressive infection which may spread to cause osteomyelitis of the temporal bone and adjacent structures. Often caused by Pseudomonas aeruginosa. The term "malignant" is an anomaly used in describing the aggressive nature of the disease rather than depicting malignancy.

Only consider ear swab for bacterial or fungal sensitivity if there is:
- Treatment failure or severe, recurrent, or chronic otitis externa.
- Ear canal occlusion due to swelling and debris, causing difficulty using topical treatments.
- Suspected spread of infection beyond external ear canal.

Reinforce self-care advice initially (Patient.info or NHS Ear Infections)
To aid recovery and reduce risk of future infection advise patient to:
- Avoid damage to external ear canal:
  - If ear wax is a problem the patient should seek advice about removing it.
  - Cotton buds or other objects should not be used to clean the ear canal.
- Keep the ears clean and dry:
  - Keep shampoo, soap, and water out of the ear when bathing and showering - use ear plugs or cotton wool (with petroleum jelly).
  - Avoid swimming/water sports for at least 7–10 days during treatment.
  - Use ear plugs and/or a tight-fighting cap when swimming to prevent recurrence.
  - Use a hair dryer (lowest heat setting) to dry the ear canal after hair washing, bathing, or swimming.
- Patients with hearing aids or those inserting earphones should limit insertion until pain and discharge (if present), have subsided.
- If allergic or has contact sensitivity to eardrops (e.g., neomycin), ear plugs, hearing aids, or earrings, avoid use, or use alternatives where available.
- Use analgesia such as paracetamol or ibuprofen and apply localised heat e.g. warm flannel for symptom and pain relief.
- Use **over-the-counter** acetic acid 2% ear spray (≥ 12 years) morning, evening, and after swimming, showering, or bathing, **for a maximum of 7 days**.

Consider cleaning, or referring to ENT for cleaning, the external auditory canal ('aural toilet'), to enable topical treatments to be applied effectively.

Treatment of acute otitis externa
Reinforce self-care advice, including use of OTC acetic acid 2% ear spray and consider 'aural toilet'.

Consider prescribing a topical antibiotic preparation **with or without** a topical corticosteroid for 7–14 days, depending on clinical judgement and symptom response.
- Medication choice is guided by ototoxicity, cost, dosing frequency, and if the tympanic membrane is intact.
- Aminoglycoside ear drops are potentially toxic and should not be used in the presence of a perforation plus discharge for >10 days, without being reassessed.
- If there is discharge, an underlying perforation is likely and should be excluded. In many cases there is no underlying perforation and ear drops can be used for longer.
- If there is a history of recurrent discharge an underlying cholesteatoma should be excluded.
- An additional topical corticosteroid preparation may be of benefit if there is significant inflammation, erythema, and oedema in the ear canal.
- Patients should be taught to properly administer ear drops (How to use ear drops or sprays | ENT UK).

Oral antibiotics are only indicated when there is evidence of spreading cellulitis (Cellulitis - Acute), signs of systemic infection or the patient is at increased risk of severe infection (immunocompromised, diabetic, a history of local radiotherapy). For children, if cellulitis is present commence oral antibiotics and arrange urgent ENT review.

Arrange follow up to reassess if:

- Symptoms not improving within 48–72 hours or not fully resolved after 2 weeks of starting initial treatment.
- Symptoms are severe and/or there is cellulitis spreading beyond the external ear canal.
- The person is immunocompromised and at risk of severe infection.
- Ear wax impaction or stenosis of the ear canal which prevents the tympanic membrane being visualised.

Treatment of chronic otitis externa

Treatment of chronic otitis externa depends on the underlying causes. Because many cases are caused by allergies or inflammatory dermatologic conditions, the most important aspect of treatment is to identify the cause and avoid further contact.

Reinforce self-care advice, consider 'aural toilet' and consider arranging an ear swab for bacterial and fungal microscopy.

If fungal infection suspected, consider:

- Topical antifungal preparation (Fungal Ear Infection Patient leaflet)
- Clioquinol and a corticosteroid topical ear drop (both anti-fungal and anti-bacterial properties)
- Over-the-counter acetic acid 2% ear spray (off label-indication, ≥12 years) for a maximum of 7 days.

If suspected bacterial infection treat as per acute otitis externa.

If no obvious bacterial or fungal infection, consider:

- Topical corticosteroid preparation.
- If symptoms persist despite topical corticosteroid, consider trial of topical anti-fungal.

Urgent referral criteria to ENT specialist:

- Otitis externa not responding to treatment, with persistent pain after 7 days,
- Chronic otitis requiring ongoing topical treatment for symptom control beyond 2–3 months.
- Diabetic and immunocompromised patients are particularly susceptible to aggressive destruction of cartilage caused by Pseudomonas aeruginosa (Necrotising Otitis Externa). If suspected, refer urgently to an ENT specialist.
- There is external ear canal occlusion due to ear discharge, swelling, or debris which is stopping topical treatment working effectively.
- There is cellulitis extending beyond the external ear canal which cannot be managed in primary care.

Topical preparations:

Medicine	Dosage	Duration	Comments
FIRST LINE			
Acetic Acid 2% spray **(Available OTC)**	One spray at least three times a day. Maximum 2-3 hourly (≥12 years)	7 days	**As effective as topical antibiotics in mild otitis externa.**
SECOND LINE			
Combined corticosteroid and **Aminoglycoside antibiotic** preparations **CONTRAINDICATED IN PERFORATED TYMPANIC MEMBRANE AND PATENT GROMMET**			
Betamethasone 0.1% and Neomycin 0.5% eye/ear/nose drops (Betnesol-N®)	Apply 2–3 drops 3–4 times a day. (Adult and child*)	7 days	Contraindicated in perforated tympanic membrane and patent grommet as contains an aminoglycoside - unless advised by a specialist for treatment of obvious infection.
Dexamethasone 0.1% with Neomycin 0.5% and acetic acid 2% (spray) **(**Previously branded as Otomize®spray.)	One spray three times a day. (Adult and children ≥2 years)	7 days	Contraindicated in perforated tympanic membrane and patent grommet as contains an aminoglycoside - unless advised by a specialist for treatment of obvious infection

Hydrocortisone 1%, neomycin sulphate 3400 units, polymyxin B sulphate 10,000 units/mL (drops: Otosporin®)	Apply 3 drops 3–4 times a day (Adults and children ≥3 years)	7 days	Contraindicated in perforated tympanic membrane and patent grommet as contains an aminoglycoside - unless advised by a specialist for treatment of obvious infection.
Dexamethasone with framycetin 0.5% and gramicidin 0.005% (Sofradex® eye/ear drops)	Apply 2-3 drops every 3-4 hours. (Adult and child *)	7 days	Contraindicated in perforated tympanic membrane and patent grommet as contains an aminoglycoside - unless advised by a specialist for treatment of obvious infection.
Hydrocortisone Acetate 1% with Gentamicin 0.3% **(Note - high cost)**	Apply 2-4 drops 4-5 times a day including a dose at bedtime. (Adult and child*)	7 days	Contraindicated in perforated tympanic membrane and patent grommet as contains an aminoglycoside - unless advised by a specialist for treatment of obvious infection.
Corticosteroid preparations Consider in positive culture resistant to both Ciprofloxacin & Gentamicin; or if it is primarily an inflammatory skin condition; and to reduce inflammation and give a break from antibiotics (excess use of ear drops with antibiotics can cause contact dermatitis).			
Prednisolone Sodium Phosphate 0.5% eye/ear drops	2-3 drops every 2-3 hours. (Adult and *)	7 days	Lower potency. Reduce frequency when relief obtained
Betamethasone 0.1% drops	2-3 drops every 2-4 hours. (Adult)	7 days	Higher potency. Reduce frequency when relief obtained
Antifungal preparations			
Clotrimazole 1% solution **(Available OTC)**	2-3 drops two to three times a day. (Adult and child *)	For at least 14 days after infection resolution	Prescribe as solution, no specific ear drop preparation available. **Counsel patient that once the symptoms have improved/disappeared, to continue treatment for 14 days.**
Combined corticosteroid and antibiotic/antifungal preparations			
Flumetasone pivalate 0.02% + clioquinol 1% (Previously branded as Locorten Vioform®.)	2-3 drops twice a day. (Adult and children ≥2 years)	7-10 days	
Other antibiotic preparations: Ciprofloxacin ear drops can be considered if there are contraindications or resistance detected on culture to aminoglycoside preparations and an antibiotic preparation is still required.			
Ciprofloxacin 0.2% (Cetraxal® UDV drops) ***Amber 2** - On specialist advice only*	Instil contents of ONE ampoule twice a day. (Adults and children ≥1year)	7 days	Previously used eye drops, now available as licensed ear drops 0.2%.
Ciprofloxacin 0.3% and dexamethasone 0.1% combination ear drops ***Amber 2** - On specialist advice only*	Apply 4 drops twice daily (Adults and children ≥1year)	7 days	
* The terms child or children are used generically to describe the entire range from infant to adolescent (1 month–17 years). An age range is specified when the dose information applies to a narrower age range than a child from 1 month–17 years. BNF			

Oral antibiotics – only if signs of spreading cellulitis.

See table below for treatment considerations in children.

Antibiotic[1]	Dosage and duration
Flucloxacillin	See Cellulitis guideline for doses (APC Cellulitis)
In penicillin allergy	
Clarithromycin[4]	
Doxycycline[2]	
Erythromycin[3,4]	

[1] See BNF for appropriate use and dosing in specific populations, for example, hepatic impairment or renal impairment, and in pregnancy and breastfeeding.
[2] Doxycycline is not suitable for pregnant women
[3] Erythromycin is preferred in women who are pregnant.
[4] Withhold statins whilst on clarithromycin/erythromycin course.

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